

Roll No.

TEC-301

B. TECH. (ECE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
ELECTRONIC DEVICES AND CIRCUITS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

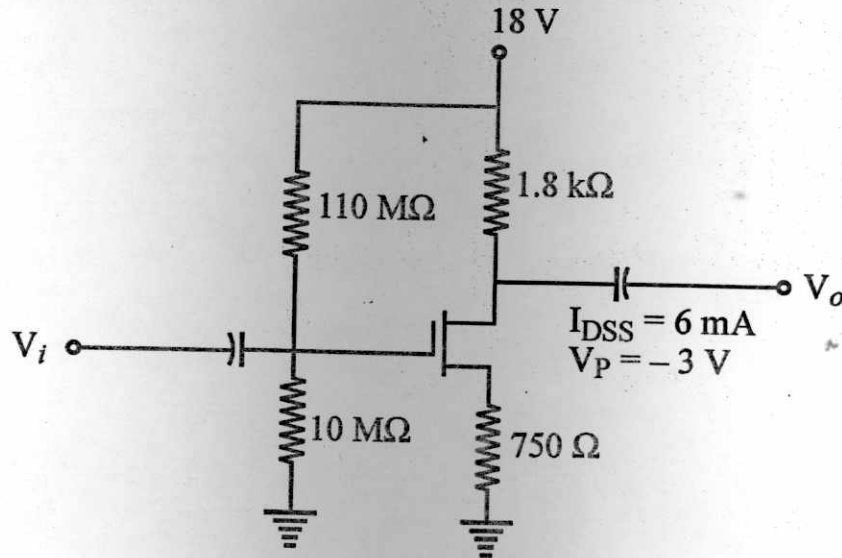
1. (a) Explain BJT as a switch and amplifier. Discuss the BJT switching timing diagram. (CO1)
- (b) Explain the following with respect to BJT : (CO1)
 - (i) Miller effect
 - (ii) Loading effect
- (c) Explain base-width modulation and its consequences in detail. (CO1)
2. (a) Discuss the transfer and output characteristics of E-only MOSFET along with the definition and expression of threshold voltage.

(CO1, CO2)

P. T. O.

(3)

- (b) Draw the circuit of Colpitt oscillator and explain its working. Derive an expression for frequency of oscillation. (CO3, CO4)
- (c) Determine I_{DQ} , V_{GSQ} and V_{DS} for the given circuit : (CO3, CO4)



5. (a) Describe the working of Class A power amplifier in detail with relevant diagram. (CO5, CO6)
- (b) Describe the working of Class B push pull power amplifier. (CO5, CO6)
- (c) Draw the circuit diagram of common-source MOSFET amplifier. Derive the expression for its voltage gain, input resistance and output resistance. (CO5, CO6)

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(iii) Total marks in each main question are *twenty*.

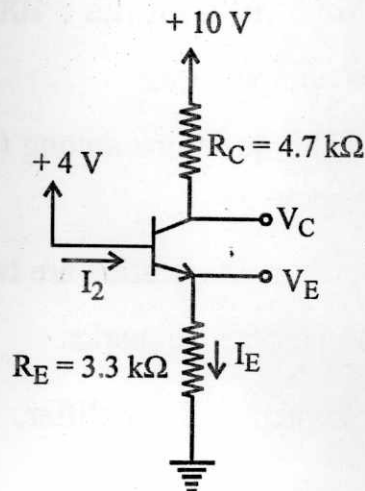
(iv) Each sub-question carries 10 marks.

1. (a) Explain BJT as a switch and amplifier. Discuss the BJT switching timing diagram. (CO1)
- (b) Explain the following with respect to BJT : (CO1)
 - (i) Miller effect
 - (ii) Loading effect
- (c) Explain base-width modulation and its consequences in detail. (CO1)
2. (a) Discuss the transfer and output characteristics of E-only MOSFET along with the definition and expression of threshold voltage.

(CO1, CO2)

P. T. O.

- (b) Discuss channel length modulation in detail, along with a proper schematic diagram. (CO1, CO2)
- (c) Explain the following terms : (CO1, CO2)
- (i) Emitter Injection Efficiency
 - (ii) Base Transport Factor
 - (iii) Punch Through
3. (a) Determine all the node voltages and branch currents for the circuit shown below : (CO3, CO5)
- Assume $\beta = 100$.

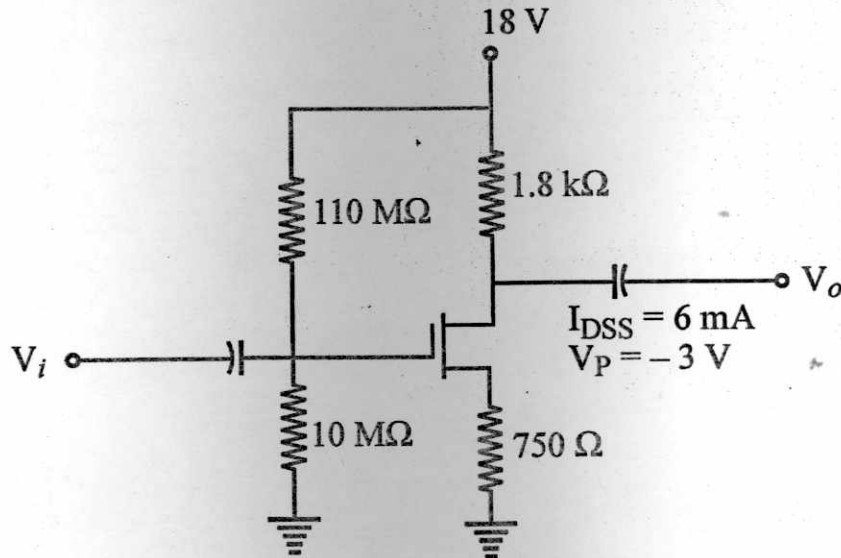


- (b) Explain the significance of threshold voltage of a MOSFET. Discuss the methods to reduce threshold voltage (V_T). (CO3, CO5)
- (c) How is oscillator different from amplifier ? What is the criterion to achieve sustained oscillations ? (CO3, CO5)
4. (a) Explain with a neat circuit diagram, the working of Wien Bridge oscillator. Derive an expression for frequency of oscillation. (CO3, CO4)

(3)

(b) Draw the circuit of Colpitt oscillator and explain its working. Derive an expression for frequency of oscillation. (CO3, CO4)

(c) Determine I_{DQ} , V_{GSQ} and V_{DS} for the given circuit : (CO3, CO4)



5. (a) Describe the working of Class A power amplifier in detail with relevant diagram. (CO5, CO6)

(b) Describe the working of Class B push pull power amplifier.

(CO5, CO6)

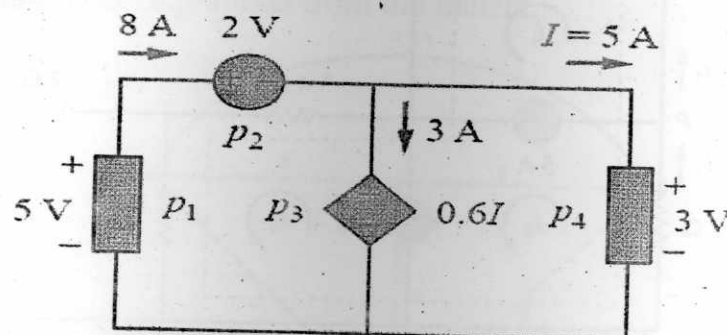
(c) Draw the circuit diagram of common-source MOSFET amplifier. Derive the expression for its voltage gain, input resistance and output resistance. (CO5, CO6)

TEC-303

B. TECH. (ECE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
NETWORK ANALYSIS AND SYNTHESIS

Time : Three Hours**Maximum Marks : 100**

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
(v) Assume parameters/values if required.
1. (a) Compute the power absorbed or supplied by each component of the circuit. show in Fig. 1 : (CO1)

**Fig. 1**

- (b) Determine v_o and i_o in the circuit shown in Figure 2.

(CO1)

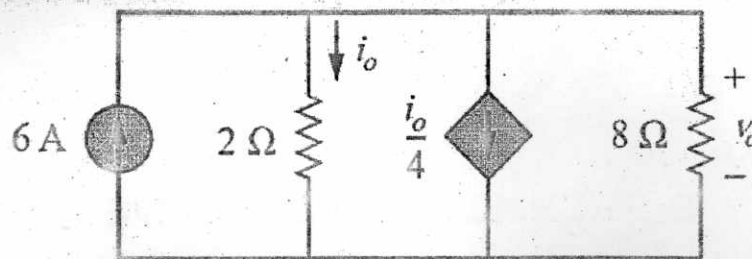


Fig. 2

- (c) Determine V_o and i in the circuit shown in Figure 3.

(CO1)

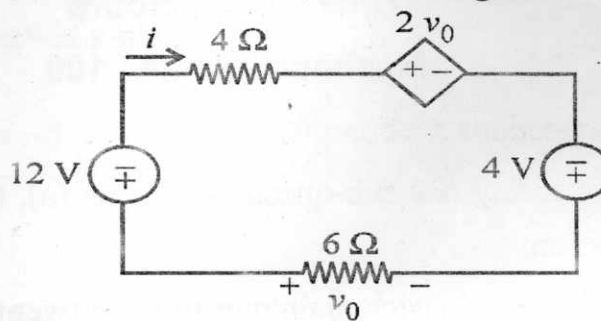


Fig. 3

2. (a) For the circuit in Figure 4, find i_1 to i_4 using mesh analysis. (CO1, CO2)

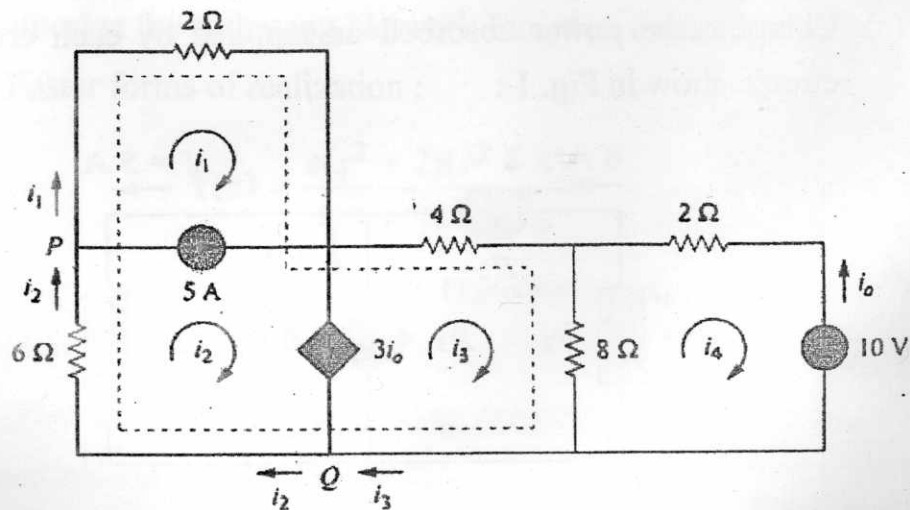


Fig. 4

OR

For the circuit shown in Figure 5 $v(t) = 100 \cos(1000t)$ volts, find the equivalent impedance for maximum power transfer.

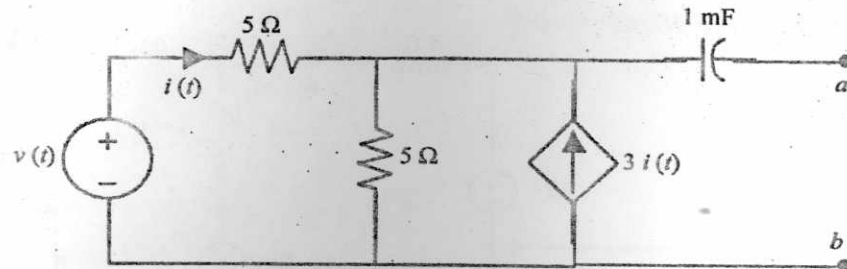


Fig. 5

- (b) Draw the directed graph of the circuit shown in Figure 6. Write the incidence matrix of the graph and hence determine the maximum number of possible trees of the graph. (CO1, CO2)

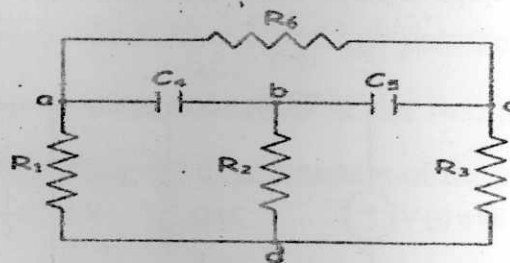


Fig. 6

- (c) Write fundamental cut set matrix and fundamental tie set matrix of any one tree of the graph shown in Figure 7 and deduce the corresponding KVL and KCL equations from the matrix. (CO1, CO2)

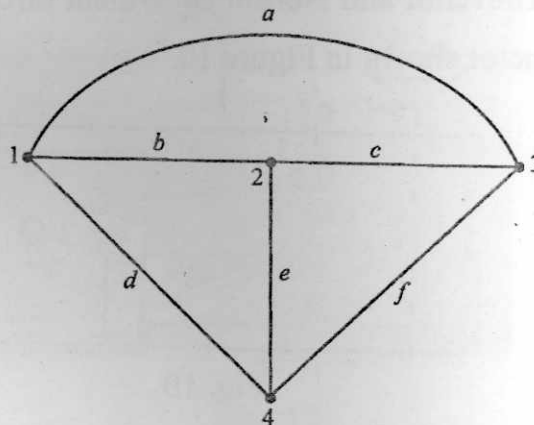


Fig. 7

P. T. O.

3. (a) At $t = 0$, switch 1 in the circuit shown in Figure 8 is closed and switch 2 is closed $4s$ later. Find $i(t)$ for $t > 0$. Calculate i for $t = 2s$ and $t = 5s$.

(CO3)

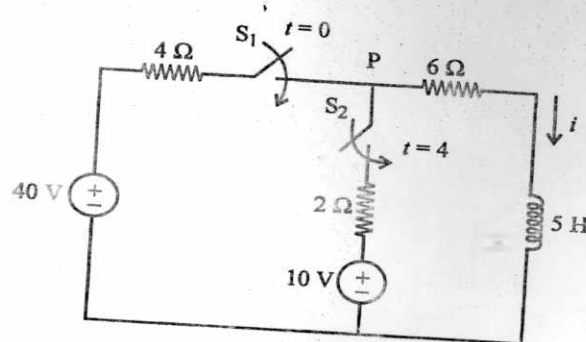


Fig. 8

- (b) In the circuit shown in Figure 9 the switch has been closed for a long time and is opened at $t = 0$. Find i and V for all time.

(CO3)

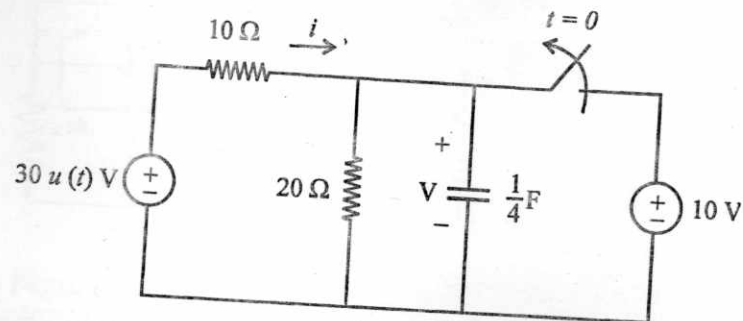


Fig. 9

- (c) Find the Thevenin and Norton equivalent circuits across the terminals of the inductor shown in Figure 10.

(CO3)

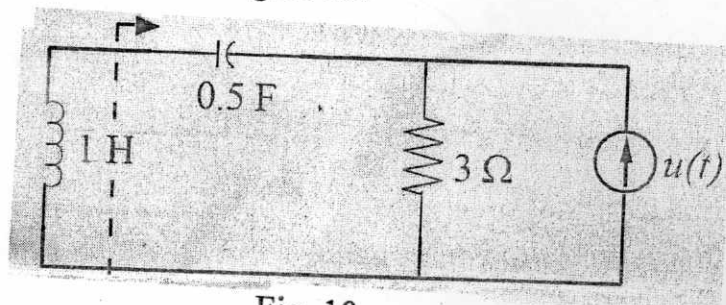


Fig. 10

4. (a) (i) Determine the Z and Y-parameters of the network shown in Figure 11 (a and b).

(CO4, CO6)

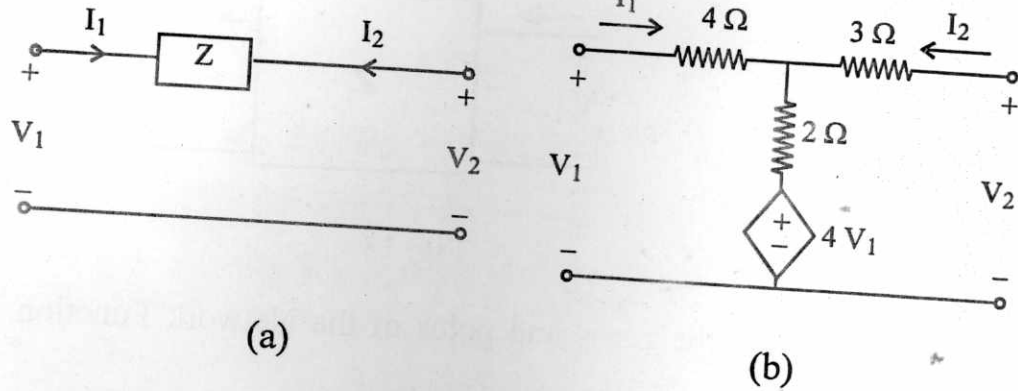


Fig. 11

- (ii) Derive the expressions of Z-parameters in terms of Transmission (A, B, C, D) parameters or Hybrid (*h*-parameters).
- (b) Identify the type of interconnection in the two port network shown in Figure 12 and determine the Z-parameters of the equivalent network.

(CO4, CO6)

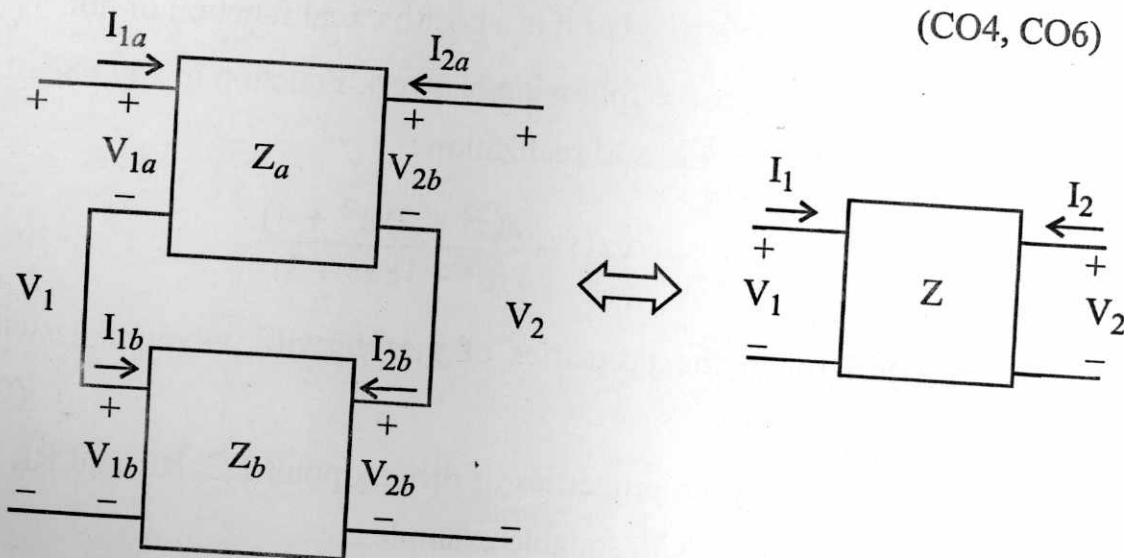


Fig. 12

- (c) Derive the expressions for the conditions of reciprocity and symmetry of a two port network shown in Figure 13. (CO4, CO6)

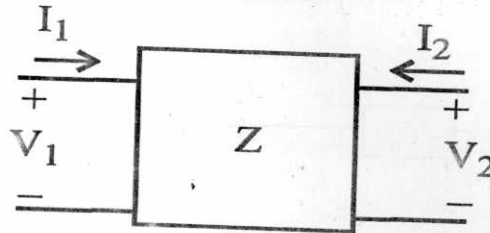


Fig. 13

5. (a) Determine the zeros and poles of the Network Function $F(s)$ and plot them in the complex s -plane : (CO5, CO6)

$$H(s) = -5 \frac{s^4 + 16s^3 + 164s^2}{(s + 32)(s^2 + 36)(s^2 + 40s + 400)}$$

- (b) (i) Given that :

(CO5, 6)

$$F(s) = \frac{s + 2}{s^2 + 3s + 2}$$

Determine whether it is a positive real function or not.

- (ii) Synthesize the following Network Function in any *one* of the Cauer or Foster forms of realization :

$$Y(s) = \frac{s(s^2 + 2)(s^2 + 4)}{(s^2 + 1)(s^2 + 3)}$$

- (c) (i) Explain the properties of the Hurwitz polynomials with suitable examples. (CO5, CO6)
- (ii) Explain the properties of driving point LC, RC and RL Impedance functions with suitable examples.

Roll No.

TEC-304

**B. TECH. (ECE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

SIGNALS AND SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Determine whether or not each of the following signals is periodic. If a signal is periodic, determine its fundamental period : (CO1)

(i) $x(t) = \cos\left(\frac{\pi}{3}t\right) + \sin\left(\frac{\pi}{4}t\right)$

(ii) $x[n] = \cos\left(\frac{\pi}{3}n\right) + \sin\left(\frac{\pi}{4}n\right)$

(b) Evaluate the following : (CO1)

(i) $\cos t \times \delta(t - \pi)$

P. T. O.

$$(ii) \int_{-\infty}^{\infty} e^{-t} \delta(2t - 2) dt$$

$$(iii) \int_{-\infty}^{\infty} e^{-t} \delta'(t) dt$$

- (c) Determine whether the following signals are energy signals, power signals, or neither : (CO1)

$$(i) x[n] = (-0.5)^n u[n]$$

$$(ii) x[n] = u[n]$$

$$(iii) x[n] = 2e^{j3n}$$

2. (a) The input $x(t)$ and the impulse response $h(t)$ of a continuous time LTI system are given by : (CO2)

$$x(t) = u(t) \text{ and } h(t) = e^{-\alpha t} u(t), \alpha > 0$$

Computer the output $y(t)$.

- (b) (i) A system has the input-output relation is given by

$$y[n] = \sum_{k=-\infty}^{n+1} x[k]. \text{ Determine whether the system is}$$

- (1) memoryless, or (2) causal. (CO2)

- (ii) Determine whether or not each of the following systems is invertible. If the system is invertible, give the inverse system :

$$(I) y(t) = 2x(t)$$

$$(II) y[n] = \sum_{k=-\infty}^n x[k]$$

(c) Verify the associative property, that is :

(CO2)

$$\{x[n] * h_1[n]\} * h_2[n] = x[n] * \{h_1[n] * h_2[n]\}$$

3. (a) Determine the complex exponential Fourier series representation for the given signals :

(CO3)

(i) $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$

(ii) $x(t) = \cos 4t + \sin 6t$

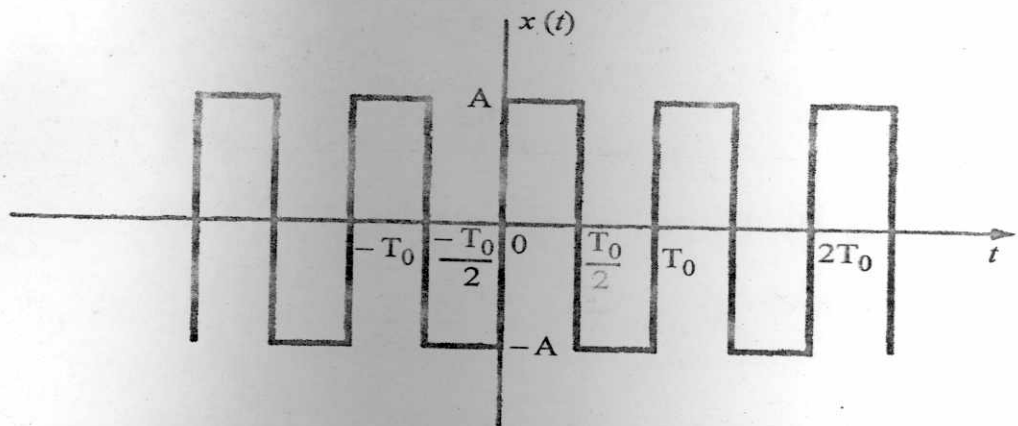
(b) Verify Parseval's theorem for the Fourier series.

(CO3)

(c) Consider the periodic square wave $x(t)$ shown in figure below.

Determine the trigonometric Fourier series of $x(t)$.

(CO3)



4. (a) Find the Fourier transform of the following signals :

(CO4)

(i) $x(t) = \text{sgn}(t)$

(ii) $x(t) = \frac{1}{a^2 + t^2}, \quad a > 0$

(iii) $x(t) = \tau \text{sinc}\left(\frac{t\tau}{2}\right)$

P. T. O.

(4)

TEC-304

(b) Verify the duality property, that is :

(CO4)

$$X(t) \xleftrightarrow{\text{F. T.}} 2\pi x(-\omega)$$

(c) Prove the frequency convolution theorem, that is :

$$x_1(t) x_2(t) \xleftrightarrow{\text{F. T.}} \frac{1}{2\pi} X_1(\omega) \cdot X_2(\omega)$$

5. (a) Find Laplace transform of the following signals : (CO5 & CO6)

(i) $x(t) = e^{-2t} \sin(2t) u(t)$

(ii) $x(t) = -te^{-2t} u(t)$

(b) Show that the final value theorem for unilateral Laplace transform is :

(CO5 & CO6)

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} s X(s)$$

(c) (i) Determine the initial value of $x(t)$ if its Laplace transform $X(s)$ is

given by $X(s) = \frac{(-2)e^{-5s}}{s(s+2)}$. (CO5 & CO6)

(ii) Determine the final value of $x(t)$ if its Laplace transform $X(s)$ is

given by $X(s) = \frac{1}{s(s-2)}$. (CO5 & CO6)

Roll No.

TMA-310

B. TECH. (ECE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

ADVANCED ENGINEERING MATHEMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Define analytic function. Find an analytic function whose real part is :

(CO1)

$$u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

(b) Use Cauchy's integral formula to evaluate :

$$\int_C \frac{3z^2 + 7z + 1}{(z-1)(z-2)} dz$$

where C is the closed curve $|z| = 3$.

(CO1)

P. T. O.

- (c) If $f(z)$ is a harmonic function of z , then show that :

(CO1)

$$\left\{ \frac{\partial}{\partial x} |f(z)| \right\}^2 + \left\{ \frac{\partial}{\partial y} |f(z)| \right\}^2 = |f'(z)|^2$$

2. (a) The following are the values for two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at various time periods (x) on individuals. Find the regression line of y on x and x on y . Also find the coefficient of correlation :

(CO1)

Time periods (x)	Diastolic blood pressure (y)
0	72
5	67
10	70
15	65
20	66

- (b) The following data representant carbon dioxide (CO_2) emission from coal-fluid boilers (in units of 1000 tons) over a period of years 1965 and 1977. The independent variables (years) has been standardized yield the following table :

(CO2)

Years (x)	CO_2 emission (y)
0	910
5	680
8	520
9	450
10	370
11	380
12	340

To fit the simple linear regression line using the above data.

- (c) From a partially destroyed lab data, following results were retrieved :
 Lines of regression are $8x = 10y - 66$ and $18y = 40x - 214$, $\sigma_x = 3$.
 Compute the following :
 $\bar{x}$, $\bar{y}$, σ_y and r for the existing data.

(CO2)

3. (a) Using Newton's method, find a real root of an equation $x \log_{10} x = 1.2$ correct to three decimals.

(CO3)

- (b) A reservoir discharging water through sluices at a depth h below the water surface, has a surface area A for various values of h as given below :

(CO3)

h (in metres)	A (in sq. metres)
10	950
11	1070
12	1200
13	1350
14	1530

If t denotes time in minutes, the rate of fall of the surface is given by :

$$\frac{dh}{dt} = -\frac{48}{A} \sqrt{h}$$

Estimate the time taken for the water level to fall from 14 to 10 m above the sluices.

- (c) Using Newton's forward interpolation formula, estimate the population for the year 1895 from the table of the population of a town in the decimal census : (CO3)

Year (x)	Population y (in thousands)
1891	46
1901	66
1911	81
1921	93
1931	101

4. (a) The velocity (v) of a particle at distance (s) from a point on its path is given by the following table : (CO4 and CO5)

Distance (s) (in metres)	Velocity (v) (m/sec)
0	47
10	58
20	64
30	65
40	61
50	52
60	38

Calculate the time taken to travel 60 metres by using Simpson's one-third rule.

- (b) A river is 80 metres wide. The depth (y) of the river at a distance (x) from bank to given by the following table : (CO4 and CO5)

x	y
0	0
10	4
20	7
30	9
40	12
50	15
60	14
70	8
80	3

Using Boole's rule, find the approximate area of cross-section of the river.

- (c) Solve $\frac{dy}{dx} = -y$ numerically from $x = 0$ to $x = 1$ using explicit forward first order Euler method with initial condition of $y(0) = 1$ and step size (h) of 0.2. Find the absolute value of error in $y(1)$ calculated using

analytical and numerical solution in percentage (%) (calculate the error using analytical solution as the basis and use three decimal places).

(CO4 and CO5)

5. (a) The number of airplane landings at an airport in a minute (X) and their probabilities are given by :

(CO6)

X_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation of X.

- (b) If the probability density function of a continuous random variable X is given by :

(CO6)

$$f(x) = \begin{cases} x^2, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

find its mean, variance and standard deviation.

(c) A random variable X has the following probability distribution : (CO6)

$X = x$	$P(X = x)$
-2	0.1
-1	k
0	0.2
1	$3k$
2	$2k$
3	0.3

Calculate the value of k and hence to compute $P(X < 2)$.

Roll No.

TEC-359

**B. TECH. (ELECTRONICS AND COMMUNICATION ENGG.
WITH SPECIALIZATION IN IOT AND B. TECH.**

**(ELECTRONICS AND COMMUNICATION ENGG. WITH
SPECIALIZATION IN ESR) (THIRD SEMESTER)**

END SEMESTER EXAMINATION, Jan., 2023

FUNDAMENTALS OF COMPUTER ORGANIZATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) List and briefly explain the main structural components of a computer.

(CO1)

(b) Explain Harvard architecture and Von-Neumann architecture with suitable block diagram. List all the advantages and disadvantages of these architectures.

(CO1)

P. T. O.

- (c) Represent signed decimal values 7, (-6), 14, (-12), 0, (-19), 1, (-42), 51 and (-31) in the following binary formats : (CO1)
- (i) Signed Magnitude
 - (ii) 1's complement
 - (iii) 2's complement
2. (a) Compare different addressing modes in terms of algorithm, advantages, disadvantages and example. (CO2)
- (b) Write a program that can evaluate the following expression: (CO2)
- $$X = (A + B) \times (C + D)$$
- using three address instructions. Assume that the processor has Load, Store, Multiply and Add instructions, and that all values fit in the registers in processor.
- (c) Explain Straight-Line sequencing and branching using suitable examples. (CO2)
3. (a) Explain briefly addition/subtraction logic unit. (CO3)
- (b) Using manual method, perform operation $A \times B$ on the 5-bit unsigned numbers $A = 10111$ and $B = 00101$. Show how same multiplication operation would be performed by sequential circuit binary multiplier. (CO3)
- (c) Design BCD adder circuit and explain briefly. (CO3)
4. (a) Differentiate between Hardwired control unit and Microporgrammed control unit. (CO4)

(3)

- (b) Explain execution of a complete instruction using suitable example. (CO4)
- (c) Explain Flynn's classification of parallel processing system. (CO4)
- 5. (a) Briefly explain different types of Read-Only Memories (ROMs). (CO5 & CO6)
- (b) Explain various mapping techniques for cache memory. (CO5 & CO6)
- (c) Compare and contrast Paging and Segmentation. (CO5 & CO6)